

**LT8365 DCDC Boost Converter****Constants**

$$U_{reference} := 1,6 \text{ V}$$

$$t_{off,min} := 115 \text{ ns}$$

**Enable Pin****Design Parameters**

$$U_{in,max} := 30 \text{ V}$$

$$U_{in,min} := 8 \text{ V}$$

$$I_{voltage\_divider,max} := 0,25 \text{ mA}$$

**Results**

$$ratio := \frac{U_{reference}}{U_{in,min}} = 0,2$$

$$R_{total} := \frac{U_{in,max}}{I_{voltage\_divider,max}} = 120 \text{ k}\Omega$$

$$R_2 := \frac{U_{reference}}{U_{in,min}} \cdot R_{total} = 24 \text{ k}\Omega$$

$$R_1 := R_{total} - R_2 = 96 \text{ k}\Omega$$

**Selected values**

$$R_1 := 100 \text{ k}\Omega$$

$$R_2 := 27 \text{ k}\Omega$$

**Results with selected values**

$$I_{voltage\_divider,max} := \frac{U_{in,max}}{R_1 + R_2} = 0,2362 \text{ mA}$$

$$U_{cutoff} := U_{reference} \cdot \frac{R_1 + R_2}{R_2} = 7,5259 \text{ V}$$

**Output Voltage****Design Parameters**

$$U_{out,max} := 330 \text{ V}$$

$$U_{out,min} := 180 \text{ V}$$

$$I_{voltage\_divider,max} := 0,25 \text{ mA}$$

**Results**

$$R_{total} := \frac{U_{out,max}}{I_{voltage\_divider,max}} = 1320 \text{ k}\Omega$$

$$R_2 := \frac{U_{reference}}{U_{out,max}} \cdot R_{total} = 6,4 \text{ k}\Omega$$

$$R_1 := R_{total} - R_2 = 1313,6 \text{ k}\Omega$$

**Selected values**

$$R_1 := 1000 \text{ k}\Omega$$

$$R_{2,fixed} := 4,81 \text{ k}\Omega$$

Parts: 4.7k, 4.87k, 4.81k, 4.99k

$$R_{2,poti} := 5,0 \text{ k}\Omega$$

Parts: 4.7k, 5k

**Results with selected values**

$$I_{voltage\_divider,max} := \frac{U_{out,max}}{R_1 + R_2} = 0,3279 \text{ mA}$$

$$U_{out,max} := U_{reference} \cdot \frac{R_1 + R_{2,fixed}}{R_{2,fixed}} = 334,2403 \text{ V}$$

$$U_{out,min} := U_{reference} \cdot \frac{R_1 + R_{2,fixed} + R_{2,poti}}{R_{2,fixed} + R_{2,poti}} = 164,6989 \text{ V}$$

**Switching frequency**

## Design Parameters

$$R_T := 130 \text{ k}\Omega$$

## Results

$$f_{switching} := \left( \frac{45,2 \text{ M}\Omega}{R_T} \right)^{\frac{1}{1,009}} \cdot 1 \text{ kHz} = 330,0109 \text{ kHz}$$

$$\text{Clear}\left(f_{switching}\right)=1$$

$$\text{Clear}\left(U_{in,min}\right)=1$$

## Output Current Capability and Inductor Selection

## Design Parameters

$$C_{laser\_driver} := 200 \text{ pF}$$

$$f_{laser\_driver} := 5 \text{ kHz}$$

$$U_{in,min} := 7,5 \text{ V}$$

$$\eta := 0,5 \quad \text{very light load means bad efficiency}$$

$$\Delta I_{SW} := 0,6 \text{ A} \quad \text{allowed output ripple current}$$

$$f_{switching} := 330 \text{ kHz}$$

## Results

$$E_{pulse} := 0,5 \cdot C_{laser\_driver} \cdot U_{out,max}^2 = 11,1717 \text{ pJ}$$

$$P_{out,avg} := E_{pulse} \cdot f_{laser\_driver} = 55,8583 \text{ mW}$$

$$I_{out,avg} := \frac{P_{out,avg}}{U_{out,max}} = 0,1671 \text{ mA}$$

$$D_{max,allowable} := 1 - t_{off,min} \cdot f_{switching} = 0,962$$

maximum allowable duty cycle

$$U_{out,single\_stage} := \frac{U_{out,max}}{3} = 111,4134 \text{ V}$$

2 Diode multiplier stage  
=> Boost output is 1/3 total output voltage

$$D_{max} := \frac{U_{out,single\_stage} - U_{in,min}}{U_{out,single\_stage}} = 0,9327$$

$$t_{off} := \frac{1}{f_{switching}} \cdot (1 - D_{max}) = 203,9904 \text{ ns}$$

Minimum switching off time is 115ns.  
Depending on the actual efficiency the  
maximum output voltage may not be  
reachable with minimum input voltage

$$I_{L,max,avg} := I_{out,avg} \cdot \frac{1}{1 - D_{max}} \cdot \frac{1}{\eta} = 4,9652 \text{ mA}$$

According to datasheet the inductor must  
handle 2.7A without saturating

$$I_{o,max} := \frac{U_{in,min}}{U_{out,max}} \cdot (1,5 \text{ A} - 0,5 \cdot \Delta I_{SW}) \cdot \eta = 13,4634 \text{ mA}$$

Max allowed output current

$$L := \frac{U_{in,min}}{\Delta I_{SW} \cdot f_{switching}} \cdot D_{max} = 35,3289 \text{ uH}$$

$$\text{Clear}(R_{total}) = 1$$

**HV Indicator LED****Design Parameters**

$$I_{LED} := 0,5 \text{ mA}$$

$$N_{Resistors} := 4$$

**Results**

$$R_{total} := \frac{U_{out,max}}{I_{LED}} = 668,4807 \text{ k}\Omega$$

$$R_{single} := \frac{R_{total}}{N_{Resistors}} = 167,1202 \text{ k}\Omega$$

$$P_{single} := I_{LED}^2 \cdot R_{single} = 0,0418 \text{ W}$$